NUST Institute of Civil Engineering (NICE-SCEE)
National University of Sciences & Technology (NUST)

CS114: Fundamentals of Programming

Class: BECE2018

Lab 07: Control Statements - I

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Lab 07: Control Statements – I

In this lab you will practice using nested if and switch statements to perform complex decision making. First you will review some programs that demonstrates the use of nested if and switch statements, then you will write your own C++ code to practice these skill.

Our aims today are to practice on:

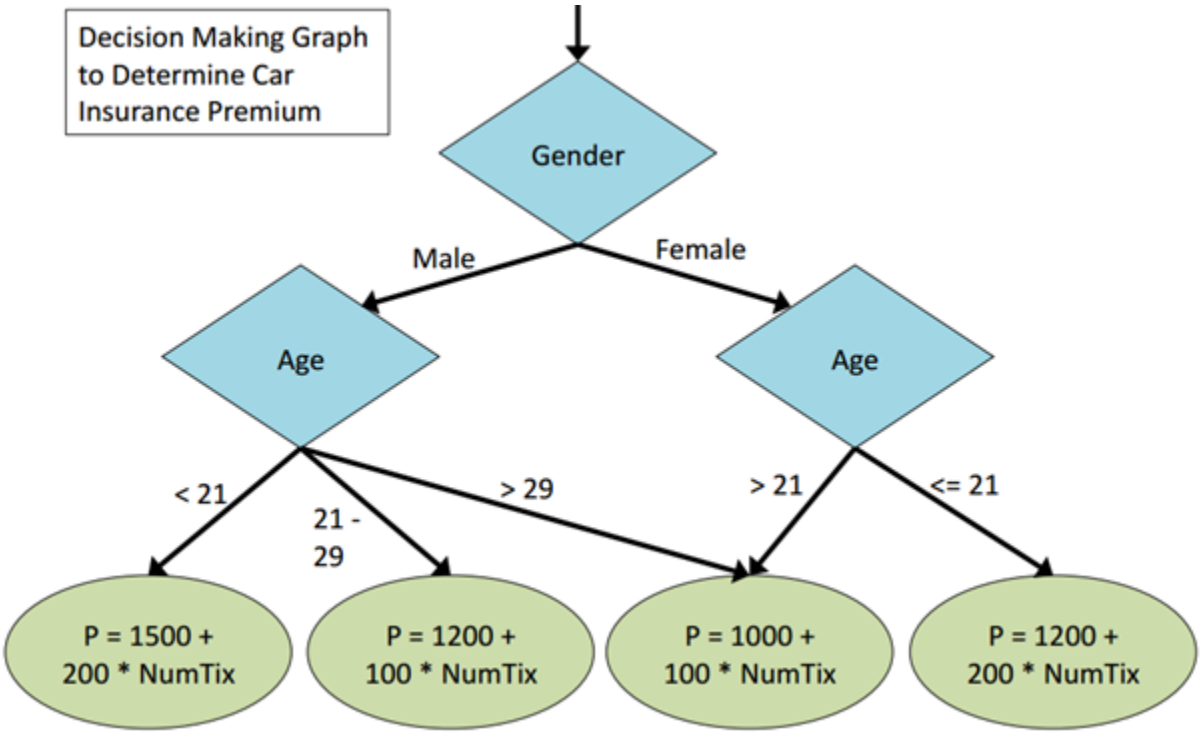
- Nested if
- switch statement

Example 1 – Nested if statements

The program simulates how a car insurance company determines your premium. The user provides the requested information and the program determines the insurance premium. The insurance premium is based on the user's gender, age, and number of traffic tickets he has received (numTix). The formula for determining the insurance premium is:

Gender	Age	Premium
Male	Under 21	1500 + 200 * numTix
Male	21 to 29	1200 + 100 * numTix
Male	Over 29	1000 + 100 * numTix
Female	Under 21	1200 + 200 * numTix
Female	21 and over	1000 + 100 * numTix

Given below is the decision making graph for the car insurance premium.



CODE

```
/* Program to calculate the car insurance premium based on the following
rules:
    - (Gender); (Age); (Annual Premium)
    - Male; Under 21; 1500 + 200 for every ticket on record
    - Male; 21 to 29; 1200 + 100 for every ticket on record
    - Male; 30 and older; 1000 + 100 for every ticket on record
    - Female; Under 21; 1200 + 200 for every ticket on record
    - Female; 21 and older; 1000 + 100 for every ticket on record
*/

#include <iostream>
using namespace std;

int main()
{

    char gender; // The customer's gender
    int age; // The customer's age


    int numTix; // Number of tickets the customer has had
    int premium; // Customer's insurance premium
```

```
// Get information from the user
// We will assume that the user always enters correct input

// Get the user's gender
cout << "Enter your gender (M/F): ";
cin >> gender;

// Get the user's age
cout << "Enter your age as an integer: ";
cin >> age;

// Get the number of tickets
cout << "How many tickets have you been issued? ";
cin >> numTix;


// Use nested if statements to calculate the insurance premium

// Outer if statement: Check the gender
if ( gender == 'M' )
{
    // The inner series of if, else if, and else statements checks
    // the age and determines the premium.
    if ( age < 21 )
    {
        premium = 1500 + 200 * numTix;
    }
    else if ( age >= 21 && age < 30 )
    {
        premium = 1200 + 100 * numTix;
    }
    else // Only other option is 30 or older
    {
        premium = 1000 + 100 * numTix;
    }
} // end if gender is male

// Matching else for the outer if statement
else // Use else because the only other option is gender == 'F'
{
    // The inner series of if, else if, and else statements tests
    // the age and determines the premium.
    if ( age < 21 )
    {
        premium = 1200 + 200 * numTix;
    }
    else // Only other option is 21 or older
    {
        premium = 1000 + 100 * numTix;
    }


} // end gender else
```

```
    // Print the result
    cout << "Your premium is $" << premium << endl;

    return 0;
}
```

The nested **if...else** statement allows you to execute a block code among many alternatives. If you are checking on the value of a single variable in nested if...else statement, it is better to use switch statement.

The **switch** statement is often faster than nested if...else (but not always). Also, the syntax of switch statement is cleaner and easier to understand.

Example 2 – switch statements

This program takes a character variable grade and displays the message to the user. Note here that is no output message and no break statement after the case 'B'. Check the behavior of the program in the given scenario.

```
#include <iostream>
using namespace std;

int main () {
   // local variable declaration:
   char grade = 'B';

   switch(grade) {
      case 'A' :
         cout << "Excellent!" << endl;
         break;
      case 'B' :
      case 'C' :
         cout << "Well done" << endl;
         break;
      case 'D' :
         cout << "You passed" << endl;
         break;
      case 'F' :
         cout << "Better try again" << endl;
         break;
      default :
        cout << "Invalid grade" << endl;
   }

   cout << "Your grade is " << grade << endl;

   return 0;
}
```

Example 3 – switch statements

```
// Program to display behavior of switch without break

#include <iostream>
using namespace std;

int main() {
    int i;
    cout << "Enter a number between 1 to 6 \n";
    cin >> i;
    switch (i) {
        case 1: cout << "1";
        case 2: cout << "2";   //execution starts at this case label
        case 3: cout << "3";
        case 4:
        case 5: cout << "5";
                break;    //execution of subsequent statements is terminated
        case 6: cout << "6";
        default: cout << "Number out of range";
    }

    cout << '\n';

    return 0;
}
```

Deliverables

1. Execute all **Examples** given in this document and take screen shots of the outputs of each example.
2. Execute all **Tasks** given in this document and take screen shots of the code and outputs of each task.

Upload single word document on LMS. Follow the naming convention for file. No late submissions will be accepted.

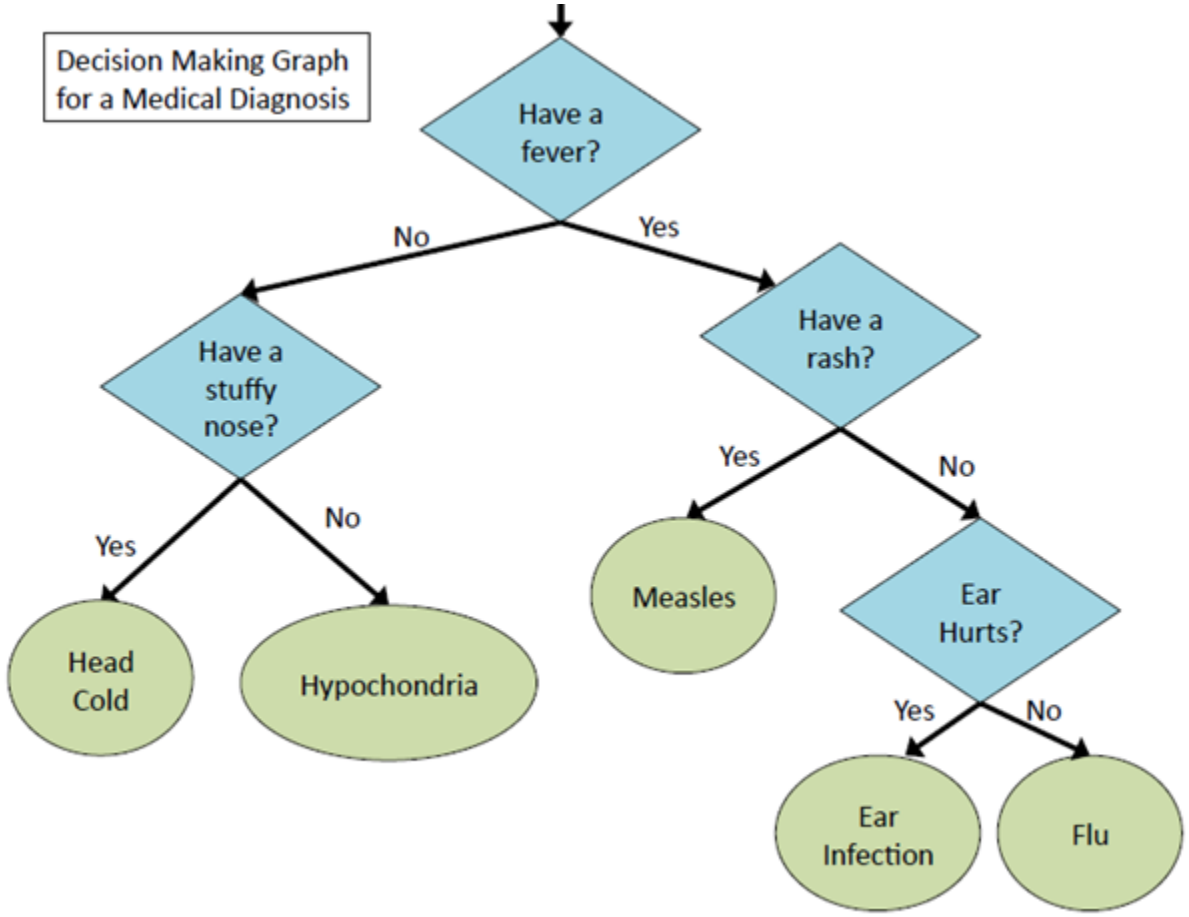
Deadline: Check deadline of the submission link on LMS.

Task 1:

You will write a simple medical diagnosis tool. Of course this is not a real medical tool, it is just an example for our program. The possible scenarios and outcomes are:

- Don't have a fever and don't have a stuffy nose: Hypocondriac
- Don't have a fever and have a stuffy nose: Head Cold
- Have a fever, don't have a rash, and ear hurts: Ear Infection
- Have a fever, don't have a rash, and ear doesn't hurd: Flu
- Have a fever and have a rash: Measles

(Decision graph given on the next page)



Task 2:

Write a program to input two numbers from user and find maximum between two numbers using switch case.

Task 3:

Write a program that takes an arithmetic operator (+, -, *, /) and two operands from a user and performs the operation on those two operands depending upon the operator entered by user and displays the output in the format **operand1 op operand2 = result.**

Task 4:

Write a program to input an alphabet and check whether it is vowel or consonant using switch case.